

Wireless Communication: Evolving to Meet Ever-Changing Needs

Wireless communication is a constantly-evolving space. Thus, it is important for the designers to be aware of the changes to the existing and new/upcoming standards so that they can incorporate the best-suited technology in the best possible way

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Evolution is the key to success.' This saying is also true in terms of wireless communication. There is a lot of competition between the different technologies associated with communication but the technology that is continuously evolving, providing a high rate of data exchange on a more secure network, is the one that can be considered to be the best.

From old standards to new, wireless communication has improved a lot over time, making communication more flexible with added reliability and a high rate of data transfer. Almost every year we see at least one new standard coming up to enhance the features associated with the older one or a new technology based on the same standard with added functionality.

Networking at 1.3Gbps

Networking technologies have been improving since the introduction of Wi-Fi standards. However, there has been a major breakthrough with the introduction of IEEE 802.11ac standard. This is an upgrade to the IEEE 802.11n standard that delivered data at 600 megabits per second (Mbps). 802.11ac incorporates wide bands, multiple antennae, beam-forming and runs at 5GHz, delivering data at the rate of 1.3Gbps. The standard also supports better penetration through walls along with longer ranges, which makes it popular amongst chip-set companies.

From application point of view, the technology would allow consumers to synchronise data instantly or transfer

Comparison between 802.11n and 802.11ac		
Technical specification	802.11n	802.11ac
Frequency	2.4, 4.9, 5GHz	5GHz
Modulation scheme	OFDM	OFDM
Channel bandwidth	20, 40MHz	20, 40, 80MHz (160MHz optional)
Nominal data rate, single stream	Up to 150Mbps (1x1, 40MHz)	Up to 433Mbps (1x1, 80MHz) Up to 867Mbps (1x1, 160MHz)
Aggregate nominal data rate, multiple streams	Up to 600Mbps (4x4, 40MHz)	Up to 1.73Gbps (4x4, 80MHz) Up to 3.47Gbps (4x4, 160MHz)
Time to stream 1.5hr HD	~ 30 min (4x4, 40MHz)	~ 15min (4x4, 80MHz)
Spectral efficiency per Gbps	400bps/Hz (4x4, 40MHz)	200bps/Hz (4x4, 80MHz)
EIRP	22-36dBm	22-29dBm
Range	12-70m indoor	12-35m indoor
Through walls	Y	Y
Non-line-of-sight	Y	Y
Worldwide availability	Y	Y, Limited in China

files quickly from one device to the other. The 802.11ac provides a more reliable network as compared to the networks with previous standards. This makes 802.11ac ideal for media streaming.

When it comes to wireless communication, apart from this, security is the major concern with the upcoming standards. The 802.11ac solves this issue by enabling high-encoding density per packet. It also works on multi-user, multiple input multiple output (MIMO), decreasing interference with an efficient spectrum usage and a great AP/client capacity.

Although 802.11ac is fast and enables easy streaming, it has certain associated shortcomings that would not only affect the designing aspect but also the working ability of the devices. For example, it claims to transfer data at a



IEEE 802.11ac router

high rate of 1.3Gbps, which is only possible at laboratory conditions. However, there is no information on the actual rate of data transfer that should be expected. Again, 802.11ac supports eight data streams, whereas the client devices support single stream and so such high-speed working is not possible unless the devices are designed to support multiple streams. The designers need to take care of the fact that 802.11ac only sup-